Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	2 October 2026
Team ID	
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Solution Requirements:

Solution requirements define what a system has to do and how well it has to do it. Functional requirements describe the features and behavior the solution offers, while non-functional requirements describe qualities such as usability, security, reliability, performance, availability and scalability.

The requirements below apply to automating standard laptop orders in ServiceNow with Flow Designer, covering the whole path from the employee's request through approval, fulfilment and delivery.

Functional Requirements:

The functional requirements of the proposed solution are listed below.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**Laptop request**	• Standard laptop catalog item available in the service catalog • Request form filled in automatically from the employee's user record • Eligibility validation and blocking of duplicate requests
FR-2	**Approval management**	• Automatic approval for pre-approved roles • Approval task delivered to the manager by email or mobile • Budget validation against the cost center • Automatic escalation of approvals pending for more than 2 days
FR-3	**Inventory and fulfilment**	• Automatic stock check in asset inventory • Laptop reserved when stock is available • Purchase order raised automatically when stock runs low • Vendor selected by region or contract price
FR-4	**Notifications and tracking**	• Notification sent at every stage (approved, shipped, delivered) • Request status shown on the employee portal
FR-5	**Asset management**	• Asset record created automatically on delivery • Asset assigned to the employee who requested it
FR-6	**Reporting**	• Dashboard showing request turnaround time • Report on automated requests and cost savings

Non-functional Requirements:

The non-functional requirements of the proposed solution are listed below.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Employees can order a standard laptop in just a few steps from a single catalog item, with clear form fields and helpful status messages.
NFR-2	**Security**	Access is controlled through ServiceNow role-based access control. Only managers can approve requests, and request and asset data can be seen only by authorized users.
NFR-3	**Reliability**	Flows run consistently. Failed steps are logged and then retried or sent to the coordinator, so that no request is ever lost.
NFR-4	**Performance**	Standard requests pass through the automated steps without manual delay, and routine flow actions finish within seconds.
NFR-5	**Availability**	Request, approval and tracking features are available during business hours, in line with the availability of the ServiceNow instance.
NFR-6	**Scalability**	The flow copes with growing request volumes, and new laptop models, roles and vendors are supported by changing configuration instead of rebuilding.

Implementation Screenshots (ServiceNow)

Captured from the project demo video: Flow Designer build and a test laptop request.

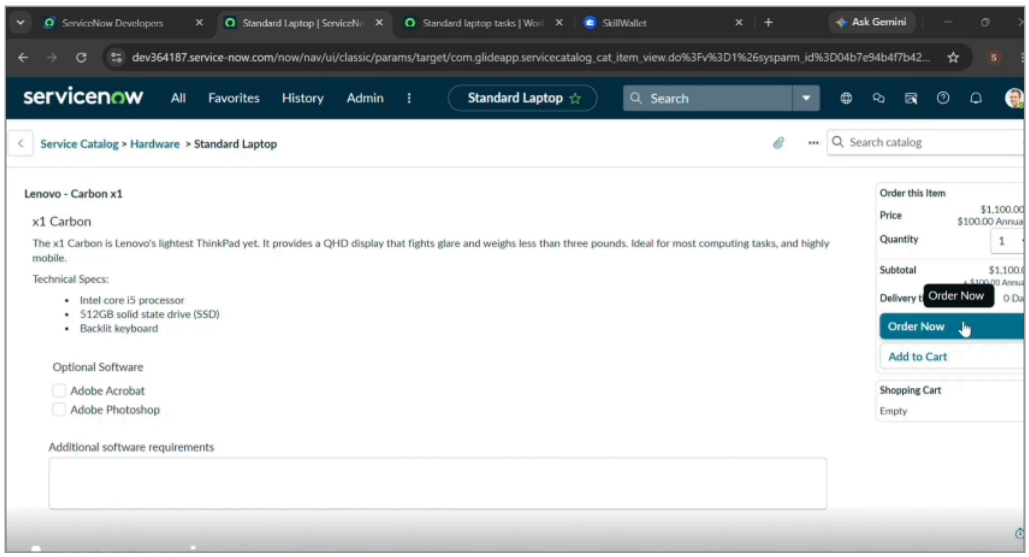


Figure 1: Standard Laptop order page with Order Now

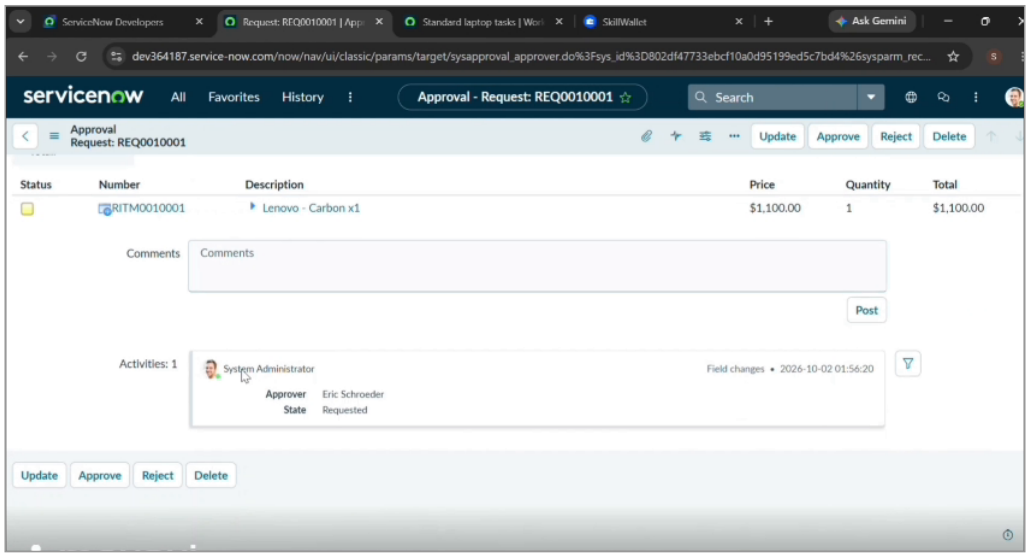


Figure 2: Approval record for REQ0010001 (Approve / Reject)

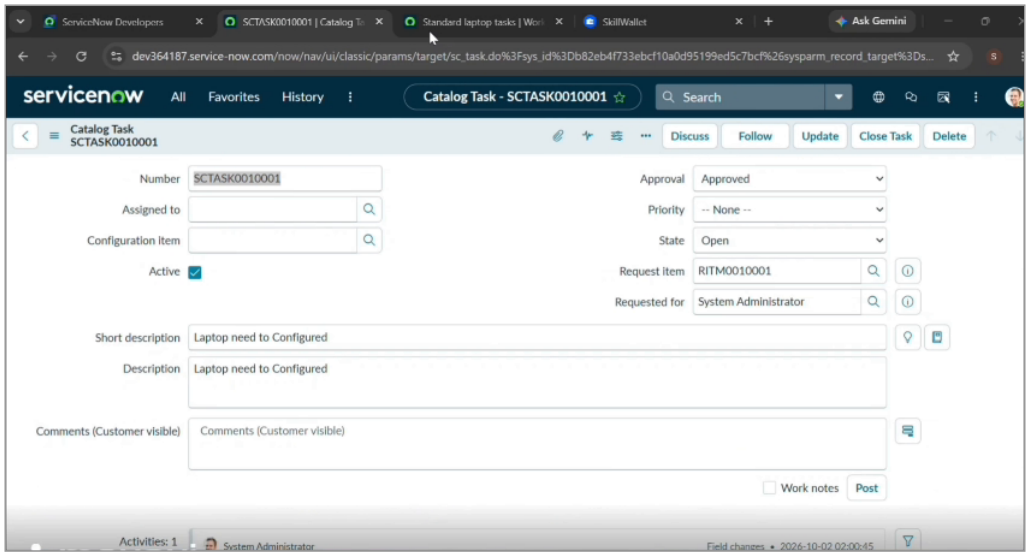


Figure 3: Catalog task SCTASK0010001 "Laptop need to Configure"